

ZHIZHGIN, Russian Federation

WMO: 224380

Lat: **65.200N**Lon: **36.817E**Elev: **55**StdP: **14.67**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.4	2.2	-5.0	4.2	-0.5	-1.6	5.1	2.7	41.2	16.9	38.3	13.4	10.6	110	0.853

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	68.5	61.6	65.6	59.7	62.9	58.1	62.7	67.0	60.5	64.5	58.7	62.3	9.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.7	79.5	64.6	58.6	73.7	63.1	56.7	68.6	60.9	28.3	67.1	26.8	64.6	25.6	62.4	70.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	32.2	28.8	26.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(5)				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.3	18.4	20.3	23.4	30.5	39.2	48.8	55.3	54.5	48.8	39.2	29.8	25.7	(6)
(7)		DBStd	14.26	8.96	8.18	7.76	6.44	6.72	6.51	5.58	4.50	4.08	4.75	5.15	6.19	(7)
(8)		HDD50	5445	980	831	825	586	340	100	14	9	68	335	605	752	(8)
(9)		HDD65	10496	1445	1251	1290	1036	798	487	304	328	485	800	1055	1217	(9)
(10)		CDD50	428	0	0	0	0	6	64	178	147	33	0	0	0	(10)
(11)		CDD65	5	0	0	0	0	0	1	3	1	0	0	0	0	(11)
(12)		CDH74	9	0	0	0	0	0	4	5	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	13.1	15.8	14.2	13.2	11.6	11.9	11.2	10.6	11.0	12.0	14.5	15.4	16.4	(14)
(15)	Precipitation	PrecAvg	14.6	0.7	0.5	0.6	0.7	1.1	1.5	1.5	2.2	1.9	1.9	1.4	0.7	(15)
(16)		PrecMax	21.9	1.3	1.2	1.3	2.0	4.2	5.9	3.6	5.4	3.8	4.0	2.9	1.3	(16)
(17)		PrecMin	9.9	0.2	0.0	0.1	0.0	0.1	0.2	0.1	0.7	0.6	0.1	0.5	0.1	(17)
(18)		PrecStd	2.5	0.3	0.3	0.4	0.5	0.8	1.0	0.9	1.1	0.9	0.8	0.6	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	34.6	38.0	48.6	61.7	72.0	73.5	69.8	61.3	51.1	41.9	35.9	(19)
(20)			MCWB	32.8	33.2	34.4	41.7	51.0	62.1	63.3	63.2	56.6	49.5	40.2	34.3	(20)
(21)		2%	DB	32.5	33.4	35.0	43.0	56.6	65.7	69.4	66.5	57.7	48.9	40.3	34.6	(21)
(22)			MCWB	31.6	32.4	32.8	38.4	49.6	57.5	62.4	61.1	54.1	47.4	39.2	33.3	(22)
(23)		5%	DB	31.4	32.4	33.9	39.7	52.8	62.3	66.6	63.7	55.7	46.9	37.8	33.6	(23)
(24)			MCWB	30.6	31.6	32.2	36.6	47.1	55.8	60.6	59.5	53.1	45.2	36.6	32.6	(24)
(25)		10%	DB	30.1	31.0	32.8	37.6	49.4	58.9	64.1	61.0	54.0	45.2	36.5	32.7	(25)
(26)			MCWB	29.5	30.2	31.4	35.1	44.9	53.3	59.3	57.6	51.7	43.6	35.3	31.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.2	33.8	34.9	42.6	53.5	63.3	65.5	65.2	57.3	49.9	40.8	35.0	(27)
(28)			MADB	33.5	34.2	36.9	48.9	58.2	70.6	70.9	67.7	60.0	50.5	41.4	35.5	(28)
(29)		2%	WB	32.0	32.7	33.3	38.9	50.2	58.9	63.3	61.9	54.8	47.6	39.2	33.7	(29)
(30)			MADB	32.3	33.2	34.4	42.5	55.4	64.7	67.9	65.4	56.6	48.7	40.4	34.1	(30)
(31)		5%	WB	30.7	31.8	32.4	37.1	47.5	56.1	61.5	59.9	53.4	45.5	36.9	32.9	(31)
(32)			MADB	31.2	32.4	33.6	39.5	52.4	61.2	65.7	63.2	55.4	46.7	37.7	33.4	(32)
(33)		10%	WB	29.4	30.2	31.6	35.5	44.9	53.5	59.6	57.9	52.1	43.9	35.5	32.0	(33)
(34)			MADB	30.0	30.9	32.8	37.2	49.3	58.5	63.4	60.7	53.8	45.0	36.4	32.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.3	7.3	7.6	7.7	8.6	9.4	8.5	7.2	5.3	3.8	4.7	5.1	(35)
(36)			MCDBR	7.3	7.0	6.1	9.8	15.0	15.9	13.1	10.8	7.6	4.4	4.7	4.7	(36)
(37)			MCWBR	7.6	7.0	5.2	7.0	10.1	10.3	8.2	7.3	5.6	4.3	4.9	4.8	(37)
(38)		5% WB	MCDBR	7.3	7.1	5.8	8.7	14.2	15.0	12.1	10.0	7.2	4.1	4.7	4.9	(38)
(39)			MCWBR	7.5	7.1	5.1	6.6	10.4	10.4	8.4	7.5	5.7	4.2	5.0	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.223	0.279	0.307	0.339	0.351	0.346	0.378	0.355	0.332	0.305	0.349	0.360	(40)	
(41)		taud	1.953	2.069	2.116	2.226	2.432	2.509	2.448	2.488	2.527	2.353	2.543	2.581	(41)	
(42)		Ebn at Noon	82	184	238	261	272	276	261	257	239	183	81	31	(42)	
(43)		Edh at Noon	7	19	30	34	31	29	30	26	21	15	7	3	(43)	
(44)	All-Sky Solar Radiation	RadAvg	29	195	679	1214	1547	1835	1661	1165	640	210	41	8	(44)	
(45)		RadStd	4	40	78	106	119	173	159	126	63	30	7	1	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page